AMARNATH S. PATEL

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EDUCATION

University of Central Florida **Undergraduate Student** **4.00 GPA**

B.S. Physics (Optics & Lasers) and Mathematics, Computer Science minor August 2025 – Expected May 2029

- **Mathematics:** Applied Linear Algebra, Linear Algebra (proof-based, in progress), Complex Analysis, Partial Differential Equations (in progress), Mathematical Methods for Physics, Ordinary Differential Equations, Honors Calculus III, Discrete Structures
- **Physics:** Quantum Information Processing, Modern Physics, Geometric Optics & Lab, Electricity & Magnetism I (in progress), Independent Research

Florida Atlantic University University coursework via FAU High School, ages 14–18 (111 credit hours) **3.66 GPA**

- Deep Learning, Data Structures & Algorithms, Computer Architecture, Computer Logic Design

RESEARCH EXPERIENCE

CELERIS — **Independent Research: From-Scratch RCWA Solver & Metalens Design Pipeline** 2026

- Built a from-scratch solver for Maxwell's equations in periodic subwavelength structures via rigorous coupled-wave analysis: 1D TE/TM gratings with Li/Liu–Fan inverse-rule factorization, full 2D-vectorial formulation for biperiodic arrays, and stable Redheffer scattering-matrix recursion.
- Validated every layer of the stack against closed-form physics, an independent method, or energy conservation — RCWA vs. transfer-matrix to 1e-6, multilayer S-matrix to 1e-15, 2D⟶ 1D degenerate reduction to 5e-12, summed diffraction efficiency 1.000000. Independently cross-validated against the grcwa and Stanford S^4 solvers to ≈ 1e-7 and reproduces canonical published metalenses (Khorasaninejad et al., *Science* 2016; Chen et al., *Nat. Nanotechnol.* 2018).
- Gradient-based (Adam) inverse design over meta-atom parameter sweeps; Monte-Carlo fabrication-tolerance analysis; Strehl, Zernike wavefront, MTF and chromatic-focal-shift analysis battery.
- Accelerated Rayleigh–Sommerfeld far-field propagation as a CUDA kernel and benchmarked it against the project's own optimized 16-core CPU path rather than a single-threaded baseline: 4.8–5.8× at 92k pillars, agreeing to machine precision. Diagnosed the kernel as memory-bound from the speedup's decline with grid size, and retracted an earlier 600–700× figure that had been measured against an unparallelized baseline. C++23/CUDA with Python bindings. *Sole-author manuscript in preparation (Computer Physics Communications).*

Undergraduate Researcher — UCF Astrophotonics Lab, CREOL August 2025 – Present

Supervisor: Dr. Stephen Eikenberry

- Measuring the wavelength-dependent complex transfer matrix of a photonic lantern by off-axis digital holography: align the interferometric bench and acquire holograms of the multimode output for each input port across the C-band (1525–1575 nm), recovering full amplitude and phase.
- Implemented the phase-retrieval and mode-decomposition analysis: FFT sideband isolation and demodulation, Butterworth low-pass, joint numerical optimization of mode-field diameter, defocus quadratic phase and field position, then decomposition onto the LP basis. Produced the per-port reconstruction-fidelity characterization for 6- and 7-port lanterns at ≈ 98% fidelity against simulated fields.
- Automated the four-instrument acquisition chain (HP 8168E tunable IR laser, Xenics Bobcat 320 InGaAs camera, DiCon GP700 fiber switch, Thorlabs MPC320 polarization control) over GPIB/VISA, RS-232 and GigE Vision, so a complete all-port × C-band sweep runs unattended with in-loop polarization optimization and saturated-frame rejection — reducing a full dataset from days of manual bench time to a single hands-off run.
- Contribute mount control and pointing automation to PolyOculus, an instrument program building photometric capability from a networked array of 8 telescopes.

Undergraduate Research Assistant — UCF Physics Department March 2026 – Present

Physics Education Research — Dr. Zhongzhou Chen (funded by NSF Award 2421299; Gates Foundation INV-076932)

- Built ESTELA, generating multi-version isomorphic exams from a structured bank of 615 problems across 29 banks, 13 topic areas and 11 question types, with auto-generated answer keys and export to four formats. Rust (Tauri 2). Materials in use by external adopters; manuscript in preparation.

Undergraduate Researcher — FAU Grant-Funded AI Safety Research Project January 2024 – March 2025

Supervisor: Tucker Hindle, Florida Atlantic University

- Benchmarked 10 statistical coherence and detection measures (GPT-2 perplexity, RoBERTa, BERT NSP, LSA, NLI, burstiness) on a 255,000-passage corpus; identified GPT-2 perplexity as the strongest discriminator (3.35× separation, 17.5 vs. 58.5) and showed BERT NSP to be uninformative.
- Built four adversarial evasion pipelines (iterative rewriting, tree-search decoding, list-branching, RL) over 3,125 generated sequences; characterized a consistent quality–evasion tradeoff. Grant-funded with HPC access; presented at the Wilkes Honors College Symposium.

TECHNICAL SKILLS

Numerical & Statistical Methods: FFT and signal processing, phase retrieval, modal decomposition, electromagnetic simulation (RCWA), gradient-based optimization, Monte-Carlo tolerance analysis, statistical discriminator benchmarking, GPU/CUDA acceleration

Programming: C/C++ (C++23), CUDA, Rust, Python (NumPy, SciPy), VHDL, Shell (Fish, Bash)

Tools: Linux, Git, CMake, Docker, Qt/PySide6, LaTeX/Typst; GPIB/VISA, RS-232, GigE Vision instrument control

HONORS & AWARDS

Barry M. Goldwater Scholarship — UCF institutional nominee	2026
Astronaut Scholarship Foundation — UCF institutional nominee	2026
Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition)	2025
Lockheed Martin Award, "Highest Level of Engineering Excellence," AEV Competition	2024
1st Place, Night Hacks Hackathon — UniUtils (course schedule generator & classroom finder)	2023